CS240 Algorithm Design and Analysis

Lecture 24

Approximation Algorithms

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Approximation Algorithms

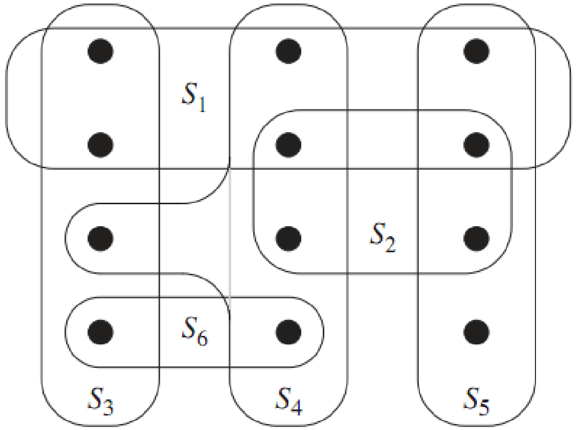
- Up to now, most of our algorithms have been exact, i.e., they find an optimal solution.
- But there are many problems for which we don't know how to find an optimal solution.
  - A key example is NP-complete problems. We don't know efficient algorithms for any NPC problem.
- Many such problems are important in practice. What do we do?
- If we can't get find the best answer, let's try for good enough.
- Approximation algorithms find an approximately optimal answer.

Approximation Ratio

- Let X be a maximization problem. Let A be an algorithm for X.
- Let a>1 be a constant.
- A is an a-approximation algorithm for X if A always returns an answer that's at least 1/a times the optimal.
  - Ex If X is max-flow, A is a 2-approx algorithm if it always returns a flow that's at least ½ the optimal.
  - The closer a is to 1, the better the approximation.
- If X is a minimization problem, A is an a-approximation algorithm for X if it always returns an answer that's at most a times larger than the optimal.
  - Ex If X is min-cut, A is a 2-approx algorithm if it always returns a cut that's at most 2 times the size of the optimal.
  - Again, the closer a is to 1, the better the approximation.

Set Covering

- **Input** A collection F of sets. Each set has a cost. The union of all the sets is X.
- **Output** A subset G of F, whose union is X.
- **Goal** Minimize the total cost of the sets in G. $$\min \sum_{S \in G} cost(S)$$



If all sets have same cost, S_3, S_4 and S_5 is a min cost set cover of X.

- Minimum cost set cover is NP-complete.
- We'll see a ln(n)-approximation algorithm, where n=|X|.

A Greedy Approximation Algorithm

- A natural greedy heuristic is to choose sets which cover points most cheaply.
  - For each set, let c be its cost, and m be the number of points it covers.
  - We want to use the set with the smallest c/m value, because this is the cheapest way to cover some new points.
- After we pick this set, remove all the points it covers. Then we consider the per unit cost of the remaining sets and again choose the cheapest.

A Greedy Approximation Algorithm

- F is the entire collection of sets. The union of these sets is X.
- Each set S in F has a cost cost(S).
- U is the set of elements of X we haven't covered yet.
- C is the set cover we eventually output.

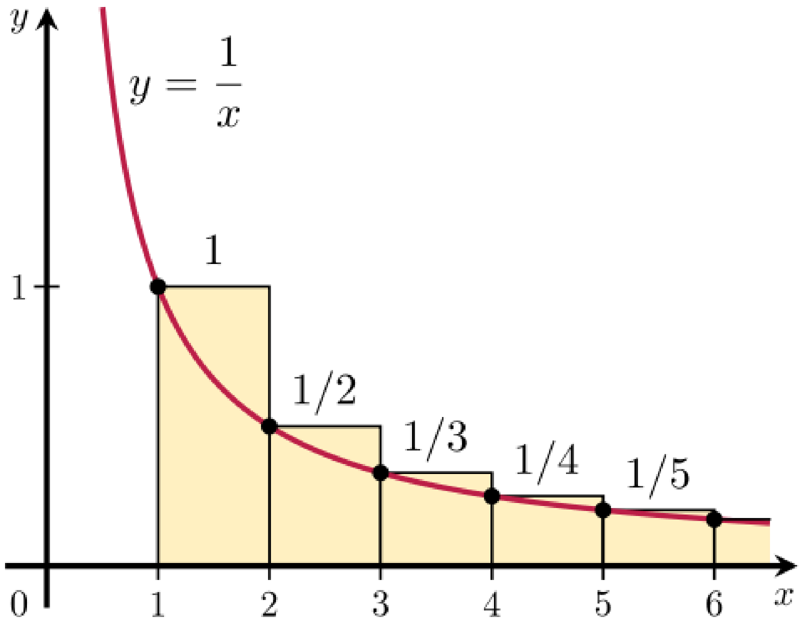
- U = X
- C = ∅
- while $U \neq \emptyset$
  - choose $S \in F - C$ with min |cost(S)|/|S∩U|
  - C = C∪{S}
  - U = U – S
- output C

Per unit cost to cover new elements.

Proof of Correctness

- We always output a set cover, because the while loop continues till X is covered.
- We'll prove the approximation ratio is at most 1+1/2+1/3+...+1/n ≈ ln(n).
  - OPT=V. The first element is charged **at most** V/n, The second element is charged **at most** $\frac{V}{(n-1)}$, For the j-th element covered by the greedy algorithm, its charge is at most $\frac{V}{n-j+1}$,
  - If the min cost of a set cover is V, our set cover costs at most ln(n)*V.
- The basic plan is to bound the cost of the set cover the algorithm outputs using the "average cost" per element.

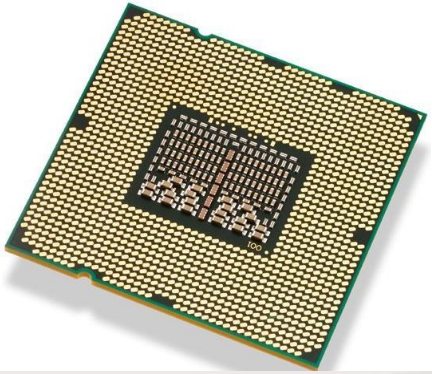
$$\frac{\text{Greedy Cost}}{\text{OPT}} \le \ln n.$$



Scheduling

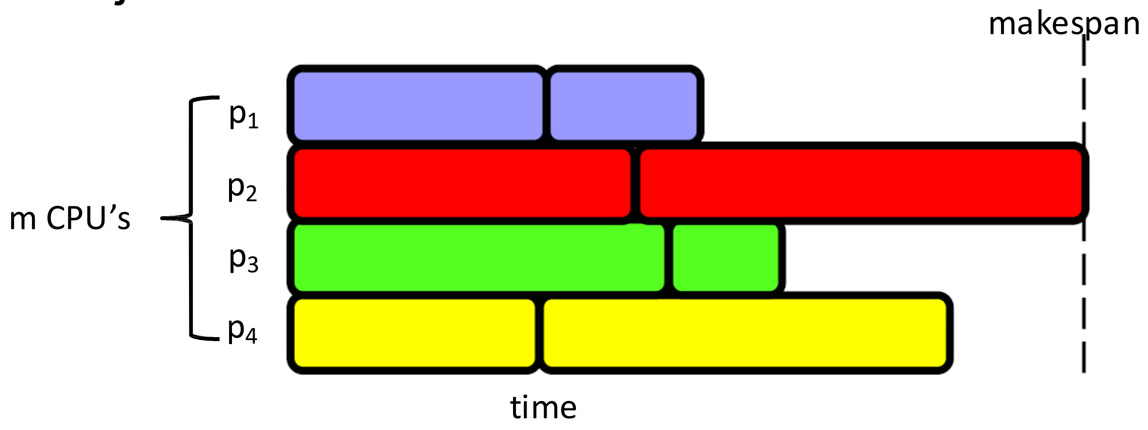
Parallel Computing and Scheduling

- Computers today are parallel.
  - Multiple processors in a system.
  - Multiple tasks for the processors to run.
- Multiprocessor scheduling is the problem of deciding which tasks to run on which processors at what time.
- Many possible objectives.
  - Throughput, fairness, energy usage.
  - Latency, i.e. finishing all jobs as fast as possible.



Makespan Scheduling

- n independent jobs.
  - Jobs have different sizes, i.e. time needed to perform job.
  - Jobs can be done in any order.
  - Any job can be done on any machine.
- m processors.
  - All have the same speed.
  - Each processors can do one job at a time.
- Assign the jobs to the processors.
- Makespan is when the last processor finishes all its jobs.
- Minimize the makespan.
  - i.e., finish all the jobs as fast as possible.



Minimizing makespan is NPC

- The decision version of scheduling is obviously in NP.
- SUBSET-SUM: given a set of numbers S and target t, is there a subset of S summing to t?
  - Ex S={1,3,8,9}. t=9, yes. t=14, no.
  - This is NP-complete. We reduce SUBSET-SUM to scheduling.
- Let (S,t) be an instance of SUBSET-SUM.
  - Let s be sum of all elements in S.
- Make a set of jobs J = SU{s-2t}, and schedule them on 2 processors.

> - **Recipe to establish NP-completeness of problem Y**
>   - Step 1. Show that Y is in NP
>   - Step 2. Choose an NP-complete problem X
>   - Step 3. Prove that $X \leq_p Y$

Minimizing makespan is NPC

- **Claim** If some subset of S sums to t, then min makespan is s-t.
- **Proof** Say S'⊆S sums to t. Schedule the jobs in S' and job s-2t on processor 1. So proc 1 finishes at time t+s-2t=s-t. Proc 2 does the jobs in S-S', so it finishes at time s-t as well.
- **Claim** If the min makespan is s-t, there exists a subset of S that sums to t.
- **Proof** Suppose WLOG proc 1 does the s-2t job. Since makespan is s-t, the other jobs proc 1 does must have total size s-t-(s-2t)=t.
- So (S,t) is yes instance of SUBSET-SUM iff makespan = s-t.
  - So SUBSET-SUM ≤ p scheduling, and scheduling is NP-complete.

Graham's List Scheduling

- Since scheduling is NPC, it's unlikely we can find the min makespan in polytime.
- List scheduling is a simple greedy algorithm.
  - Finds a schedule with makespan at most twice the minimum.
  - A 2-approximation.
- If there are n tasks and m processors, list scheduling only takes O(n log n + m) time.
  - Compare this to n! C(n+m-1, m-1) time to try all possible schedules and pick the best.

Graham's List Scheduling

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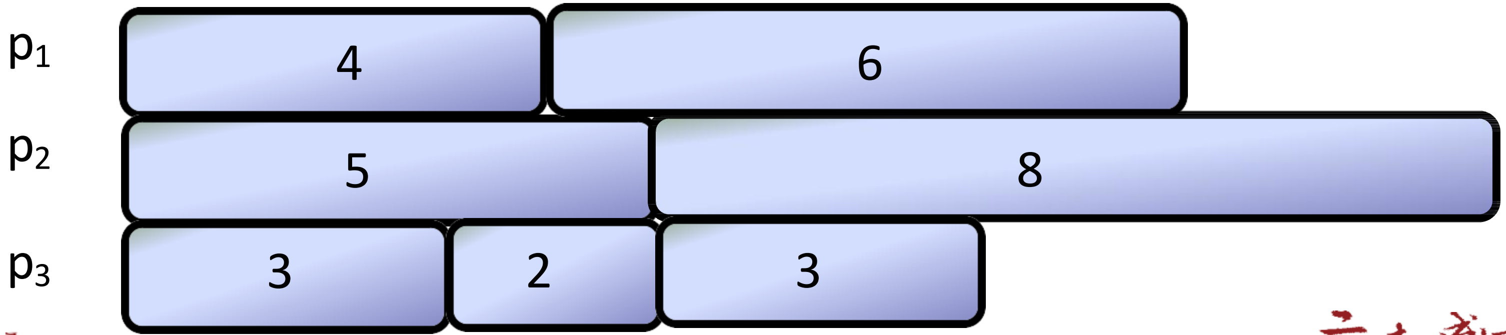
- List the jobs in any order.
- As long as there are unfinished jobs.
  - If any processor doesn't have a job now, give it the next job in the list.
- Example
- 3 processors. The jobs have length 2, 3, 3, 4, 5, 6, 8.
- List them in any order. Say 4, 5, 3, 2, 6, 8, 3.
- Initially, no proc has a job. Give first 3 jobs to the 3 procs.
- At time 3, proc 3 is done. Give it next job in list, 2. Give
- At time 4, proc 1 is done. it next job in list, 6.
- At time 5, both 1, 3 are done. Give them next jobs in list, 8,3.
- Everybody finishes by time 13.
  - The makespan of this schedule is 13.



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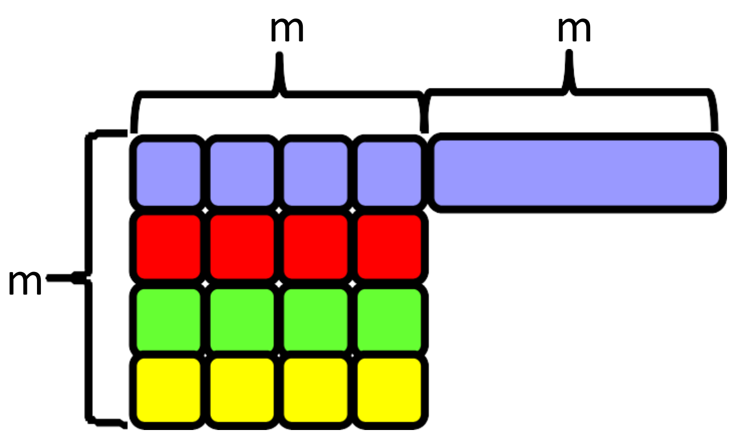
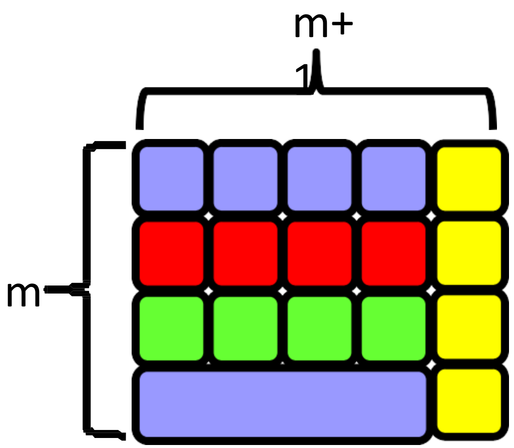
The Worst Case for LS

- How badly can list scheduling do compared to optimal?
- Say there are m^2 jobs with length 1, and one job with length m.
  - Suppose they're listed in the order 1,1,1,...,1,m.
  - LS has makespan 2m. Optimal makespan is m+1.
  - makespan(LS) / makespan(opt) = 2m/(m+1) ≈ 2.
- This is worst possible case for list scheduling.



makespan(LS) = 2m

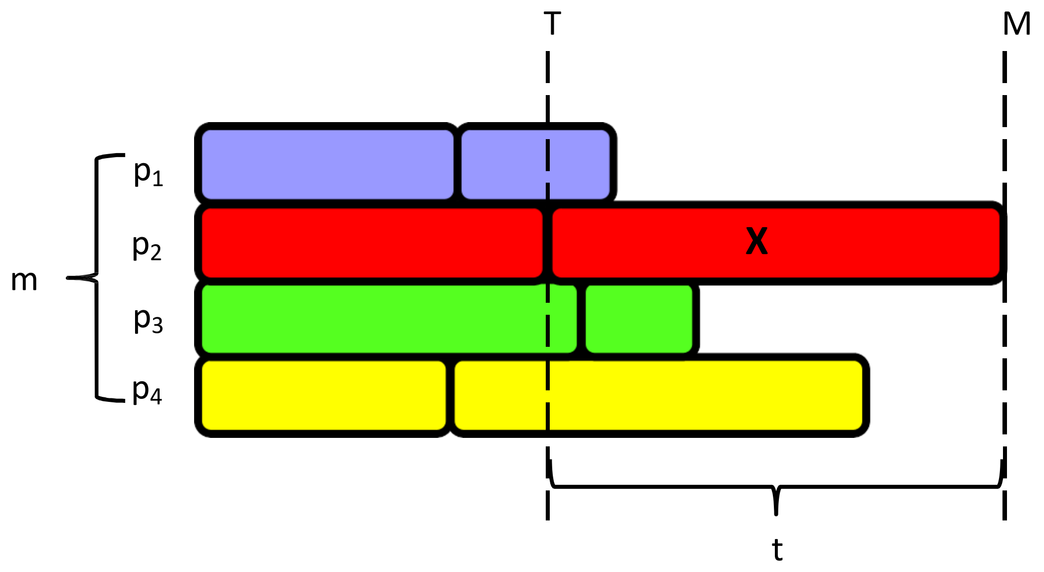


makespan(opt) = m+1

LS is a 2-approximation

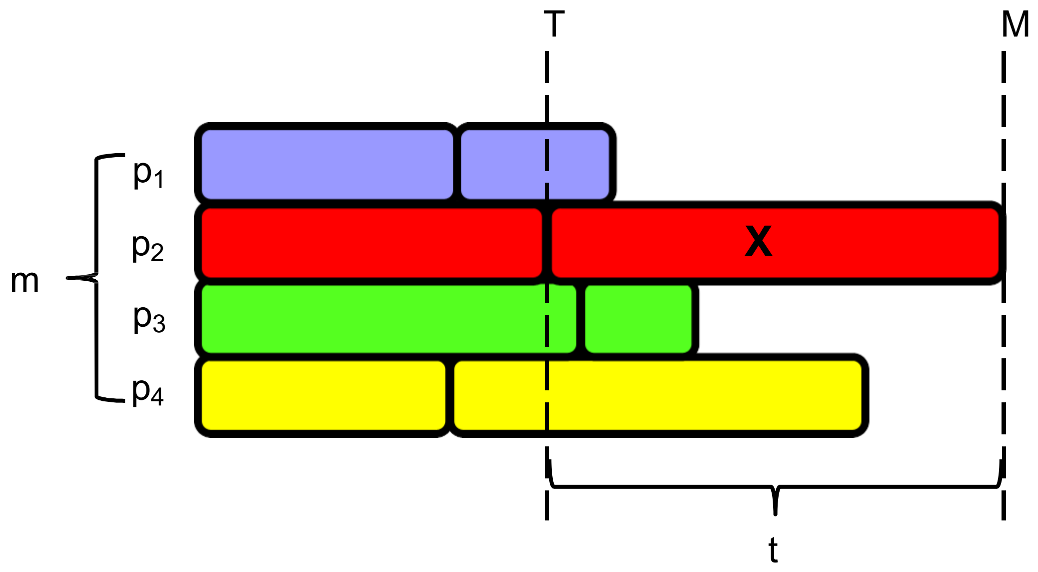
- Next, we prove LS always gives a schedule at most twice the optimal.
- Suppose LS gives makespan of M.
- Let the optimal schedule have makespan M*.
- We prove that $M \le 2M^*$.

LS is a 2-approximation



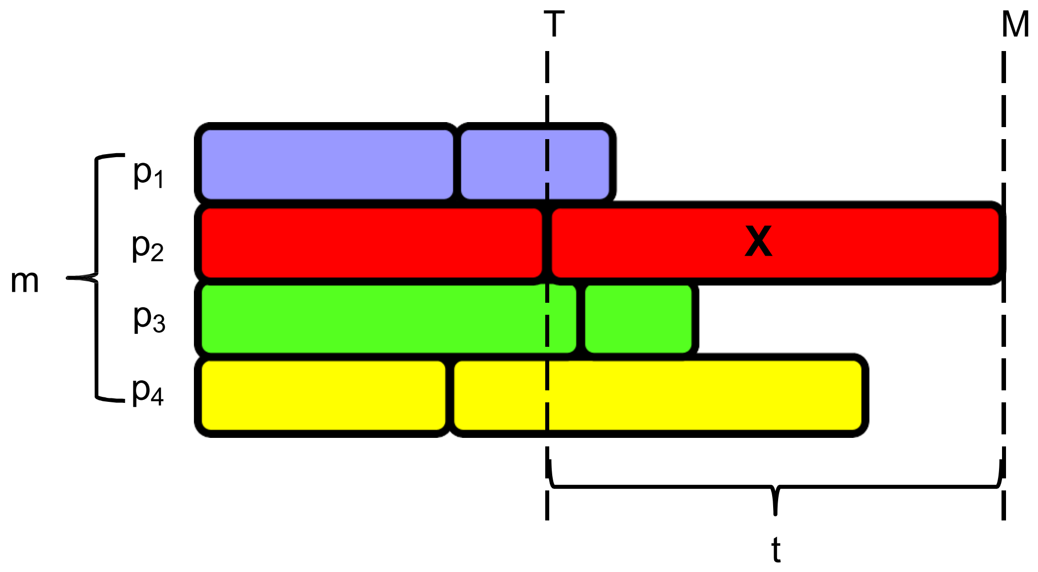
- The picture above is the schedule produced by list scheduling.
- Consider task X that finishes last.
  - Say X starts at time T, and has length t.
- **Claim 1** $M^* \geq t$.
  - In any schedule, X has to run on some process.
  - Since X takes t time, every schedule, including the opt, takes ≥ t time.

LS is a 2-approximation



- **Claim 2** $M^* \geq T$.
  - Up to time T, no processor is ever idle.
    - Up to T, there's always some unfinished job.
    - As soon as a processor finishes one job, it's assigned another one.
  - So at time T, each processor completed T units of work.
  - So total amount of work in all the jobs is $\geq mT$. Up to T: mT
  - In the opt schedule, m processors complete at most m units of work per time unit.
  - So length of opt schedule is $\geq$ (total work)/m $\geq mT/m = T$.

LS is a 2-approximation



- From Claims 1 and 2, we have M* ≥ t and M* ≥ T.
- So M* ≥ max(T,t).
- M = T + t, because X is last job to finish.
- So M/M* ≤ (T+t)/max(T,t) ≤ 2.

$$\frac{M}{OPT} \leq \frac{T+t}{\max(T,t)} \leq 2.$$

LPT Scheduling

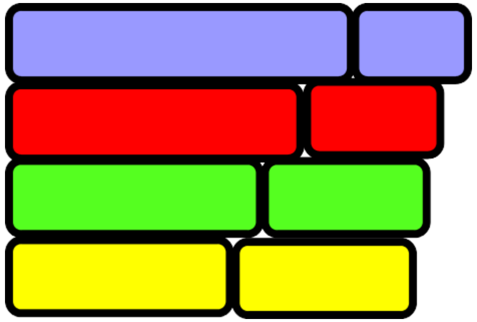
- Worst case for LS occurred when longest job was scheduled last.
  - Large jobs are “dangerous” at end.
  - Let’s try to schedule longest jobs first.
- Longest processing time (LPT) schedule is just like list scheduling, except it first sorts tasks by nonincreasing order of size.
- **Ex** For three processors and tasks with sizes 2, 3, 3, 4, 5, 6, 8, LPT first sorts the jobs as 8,6,5,4,3,3,2. Then it assigns p1 tasks 8,3, p2 tasks 6,3, p3 tasks 5,4,2, for a makespan of 11.
- LPT has an approximation ratio of 4/3.

LPT is a 4/3-approximation

- **Thm** Suppose the optimal makespan is M*, and LPT produces a schedule with makespan M. Then $M \leq 4/3\ M^*$.
- Let X be the last job to finish. Assume it starts at time T and has size t.
- Assume WLOG that X is the last job to start.
  - If not, then say Y starts after T.
  - Y finishes before T+t. So we can remove Y without increasing the makespan.
- **Cor 1** X is the smallest job.
  - X is the last job to start, so due to LPT scheduling it's the smallest.

LPT is a 4/3-approximation

- **Claim 1** LPT's makespan = $T+t \le M^*+t$.
  - As in LS, no processor is idle up to time T, so $M^* \ge T$.
- **Case 1** $t \le M^*/3$.
  - Then LPT's makespan $\le M^* + t \le M^* + M^*/3 = 4/3\ M^*$.
- **Case 2** $t > M^*/3$.
  - Since X is the smallest task, all tasks have size $> M^*/3$.
  - So the optimal schedule has at most 2 tasks per processor. So $n \le 2m$.
  - If $1 \le n \le m$, then LPT and optimal schedule both put one task per processor.
  - If $m < n \le 2m$, then optimal schedule is to put tasks in nonincreasing order on processors 1,...,m, then on m,...,1.
    - LPT also schedules tasks this way, so it's optimal.



LS VS. LPT

- LPT gives better approximation ratio, has same running time. Why bother with LS?
- LS is online.
  - Imagine the jobs are coming one by one.
    - LS just puts them on any idle computer.
- LPT is offline
  - It needs to know all the jobs that will ever arrive, in order to sort them.
- In a realistic parallel computation, you get jobs on the fly.
  - Online is more realistic.
  - LS is usually more useful.